

The Universal Warmup Path: Automatic Preconditioner Selection for HMC

Junpeng Lao

26 July 2026

Abstract

Euclidean Hamiltonian Monte Carlo (HMC) warmup must choose a step size and constant preconditioner from limited, nonstationary draws. Standard warmup follows a fixed schedule and generally requires the preconditioner structure to be specified in advance. We present a multi-chain controller that starts diagonal and, at dimension-derived window endpoints, selects between diagonal and low-rank-plus-diagonal inverse mass matrices and chooses the retained rank, subject to dimension and sample-support caps. When evidence is inconclusive, the controller gathers another scheduled window. Persistent within-/between-chain disagreement means draws do not support treating one constant preconditioner as an adequate global description; the controller retains its within-region matrix and advises a population or tempering method for regional exploration. Poor held-out score–position linearity advises reparameterization. The controller selected low rank in every evaluated headline benchmark NUTS run (12/12). Geometric-mean pooled ESS-per-gradient ratios relative to the prespecified Fisher low-rank warmup and Welford diagonal warmup baselines were respectively 2.451 and 22.572 on the synthetic ill-conditioned Gaussian benchmark, and 1.951 and 6.264 on the German-credit Bayesian logistic-regression posterior; all compared runs passed the post-warmup quality check. This method unifies common HMC warmup heuristics in one evidence-driven controller, reducing manual choices and turning warning signals into actionable guidance.

1 Introduction

Local curvature determines how finely HMC must integrate and how efficiently it can move within a region. Global posterior structure determines whether those locally efficient transitions explore the target adequately [Duane et al., 1987, Neal, 2011]. Euclidean HMC therefore needs a step size and a constant mass matrix, while NUTS also adapts trajectory length [Hoffman and Gelman, 2014]. Warmup must estimate the step size and preconditioner from limited, nonstationary draws while keeping conclusions conditioned on the regions represented in those draws.

Standard window adaptation uses a fixed schedule and generally asks the user to choose a diagonal or dense mass-matrix structure before execution. The controller studied here instead begins diagonal, schedules its evidence checks from dimension, chain count, and gradient budget, and can promote from diagonal to a low-rank-plus-diagonal constant inverse mass matrix. Without a supported promotion it remains diagonal. Inconclusive evidence causes it to gather

another scheduled window. Persistent within-/between-chain disagreement retains the applicable within-region matrix and produces advice to use a population or tempering method for regional exploration. Poor held-out score–position linearity instead produces reparameterization advice.

Previous work by Bales et al. [2019] constructs and selects for deployment diagonal, dense, and Hessian-informed low-rank constant Euclidean preconditioners from one chain’s warmup window, using an 80/20 held-out conditioning criterion for leapfrog stability. Our finite multi-chain within/between evidence instead gates estimator and retained-rank eligibility, yielding another window if inconclusive or a population handoff under persistent disagreement.

For the formal analysis, a indexes candidate preconditioner estimators and $m = 1, \dots, M$ indexes chains. Each estimator maps statistics from one current M -chain window to a candidate constant inverse mass matrix, whose population target is

$$G_a^* = T_a(\pi).$$

They use the same window but different summaries: diagonal scale, pooled per-chain-centered within-chain information, or between-chain means. The latter two yield low-rank corrections and may share a label while targeting different population objects. In the theorem sections, a “route” means a selected estimator. During a stable route episode, the population preconditioner map may then attract the iterate to G_a^* .

Four questions remain distinct throughout. *Attraction* asks whether an estimator’s fixed population update contracts. *Structural identification* asks whether the transcript resolves a rank or subspace. *Representativeness* asks whether the transcript has missed important regions. *Utility* asks whether an identified structure improves compute-adjusted sampling. None implies the next. Here estimator or preconditioner identification means recovering covariance structure, rank, or subspace from a warmup transcript; it is unrelated to causal identification or statistical-model identifiability.

“Universal” in the title refers to using one automatic controller across the evaluated HMC-family kernels. Under the declared constant-preconditioner adequacy criterion, score–position nonlinearity produces reparameterization advice, while persistent within-/between-chain disagreement produces advice to use a population or tempering method for regional exploration. In either case, the selected within-region preconditioner remains available for local HMC transitions.

Our contributions are a route-indexed attractor theorem with explicit finite-window error terms; a Markov-transcript uncertainty construction and its operator consequences; a finite-horizon local-transcript information bound; and an automatic HMC warmup implementation whose diagnostic threshold tests keep identification, consistency, and utility separate. The controlled Gaussian-mixture study holds the marginal spectrum fixed while the recorded within/between evidence and selected estimator change, exposing both the value and the limits of the diagnostics. Together, the fixed-geometry benchmarks test efficiency after selection, while the mixture sweep tests whether selection changes with the evidence. The empirical corpus compares the controller with prespecified Fisher low-rank warmup and Welford diagonal warmup baselines, and with a historical shared-step implementation.

Section 2 fixes the mass-matrix convention, and Section 3 gives the practical decision tree. Sections 4–6 develop the conditional guarantees and limits. Section 7 reports the empirical comparisons; implementation, related work, and scope follow, with proofs and supporting studies in the appendices.

2 Mass-matrix convention and candidate estimators

Let $\pi(x) \propto \exp\{\mathcal{L}(x)\}$ on $\mathbb{R}^d$. We use the *inverse-mass* convention

$$p \sim \mathcal{N}(0, G^{-1}), \quad K(p) = \frac{1}{2}p^\top G p. \quad (1)$$

Many Bayesian inference libraries, including NumPyro and BlackJAX, use `inverse_mass_matrix` for $G = M^{-1}$, where M is the HMC mass matrix. For $x \sim \mathcal{N}(\mu, \Sigma)$, Hamilton’s equations give $\ddot{x} = -G\Sigma^{-1}(x - \mu)$, so the covariance-matching choice is $G^\star = \Sigma$. In this convention the inverse mass matrix equals the target covariance for a Gaussian. All positions, scores, covariance references, and geometry statements in this paper use the unconstrained coordinates supplied to HMC, after any transformation from constrained model parameters.

For a general target with finite second moments, define the posterior covariance reference

$$\Sigma_\pi = \text{Cov}_\pi(X).$$

It is a common reference for the HMC preconditioners studied here, not a common estimator target: a Fisher, masked, regularized, or finite-rank estimator need not target Σ_π .

For an estimator a , let $G_a^\star = T_a(\pi)$ and define the affine-invariant residual

$$R_a(G) = G^{-1/2}G_a^\star G^{-1/2}.$$

The tangent from G toward $G_a^\star$ is represented by $\log R_a(G)$ in the whitened frame. This is not interchangeable with the log-Euclidean difference $\log G_a^\star - \log G$ unless the matrices commute. For the recursion below we work in a bounded SPD chart $\theta = \log G$; the per-estimator moment-to-preconditioner map is part of T_a , not silently identified with either matrix logarithm.

Classical window adaptation updates step size in fast and slow intervals, while learning covariance only in expanding, memoryless slow intervals [Carpenter et al., 2017, Stan Development Team, 2026]. Fisher-divergence preconditioning supplies an estimator of a constant mass matrix, or affine preconditioner, from warmup draws and scores in diagonal, dense, or low-rank-plus-diagonal form [Seyboldt et al., 2026]. Its mass-matrix estimation and dual-averaging step-size adaptation are separate, and the paper also specifies its own warmup schedule. Our controller must decide not only which estimator to use, but when the transcript makes that estimator eligible.

3 Automatic HMC warmup controller and evaluation protocol

The evaluated implementation takes a target log density and score, dimension d , chain count M , an HMC-family kernel, and total gradient budget $\mathcal{B}$. Initial positions are supplied by the caller; between-chain evidence is informative to the extent that these starts and the subsequent warmup expose distinct regions. Before execution the controller constructs the complete decision schedule from d , M , the kernel cost, and $\mathcal{B}$. It starts with a diagonal inverse mass matrix while adapting step size. For NUTS it assigns $\lfloor \mathcal{B}/(20M) \rfloor$ transitions to each chain, using 20 as a conservative gradients-per-transition allowance; a fixed- L kernel uses its declared integration length in place of 20. The dimension-derived rank capacity and pooled support floor are

$$k_{\text{cap}} = \min\{50, \max(\lfloor d/2 \rfloor, 1)\}, \quad N_{\text{min}} = 8(k_{\text{cap}} + 1).$$

After a one-step initialization window, the first evidence-bearing window has $n_1 = \lceil N_{\min}/M \rceil$ transitions per chain. Mass-matrix windows then have nominal $1.5\times$ growth. When another grown window would not fit before the step-size-only phase, the last slow window instead absorbs the remaining mass-matrix transitions. The final 15% adapts step size only. These boundaries are fixed before sampling begins.

At each prescribed mass-matrix-window endpoint, the controller pools the current per-chain buffer across the M chains, computes within-chain and between-chain evidence, runs the implemented conjunction of diagnostic threshold tests, and resets the buffer. The conjunction combines the within-chain and between-means structural tests, the R^2 constant-preconditioner-fixability check, and enough remaining budget for the selected rank and step-size readaptation. These structural tests are denoted W and T, respectively, in the implementation. The held-out score-position linearity check asks whether the observed local geometry is adequately approximated by a constant linear preconditioner; poor fit motivates reparameterization. Persistent within-/between-chain disagreement instead motivates a population or tempering method for regional exploration. The controller treats both as action signals conditioned on the observed warmup evidence.

A supported decision promotes the diagonal matrix to a low-rank-plus-diagonal matrix. The within-chain estimator constructs its low-rank correction from pooled, per-chain-centered information; the between-means estimator uses a correction based on the chain means. If no promotion is supported, the final choice remains diagonal. Inconclusive evidence retains the diagonal matrix and advances to the next larger scheduled window. Persistent within-/between-chain disagreement can instead indicate that the draws do not support treating one constant preconditioner as adequate across the observed regions or as evidence of global exploration. In that case the controller retains the applicable within-region matrix and reports advice to use a population or tempering method for regional exploration. Poor held-out score-position linearity separately supports reparameterization advice. Decisions are evaluated at endpoints, and every window runs to its prescribed fixed boundary.

On completion, `parameters` contains the frozen `step_size` and `inverse_mass_matrix`. The controller verdict reports `route` and `effective_rank`; its flags report `metric_scope`, `observed_ensemble_evidence`, and `handoff`. The selected constant preconditioner remains usable for within-region HMC transitions, while a population or tempering method can support regional exploration. Advisory outputs are recomputed at later endpoints and may clear while the scan continues.

Estimator, window schedule, mass-matrix-buffer policy, and step-size controller are independent engineering choices. The Fisher-HMC reference prescription in this paper is specifically the 30/55/15 schedule of [Seyboldt et al. \[2026\]](#): it uses $L = 10$ and then $L = 80$ refresh/memory periods, periodically wipes older draws, and reinitializes step size once at the start of phase 2. Its constant-mass-matrix estimator and dual-averaging step-size adaptation are distinct. This description is specific to the cited Fisher-HMC prescription. Stan’s expanding slow windows are successively doubled in the version 2.39 CmdStan User’s Guide [[Stan Development Team, 2026](#)].

The empirical schedule study estimates performance of complete configurations because schedule, buffer policy, and restart behavior vary together. Its limited reseed ablation reports the observed continuous-versus-reseed comparison. Figure 5 in Appendix D records the schedule prescriptions and one executed trace.

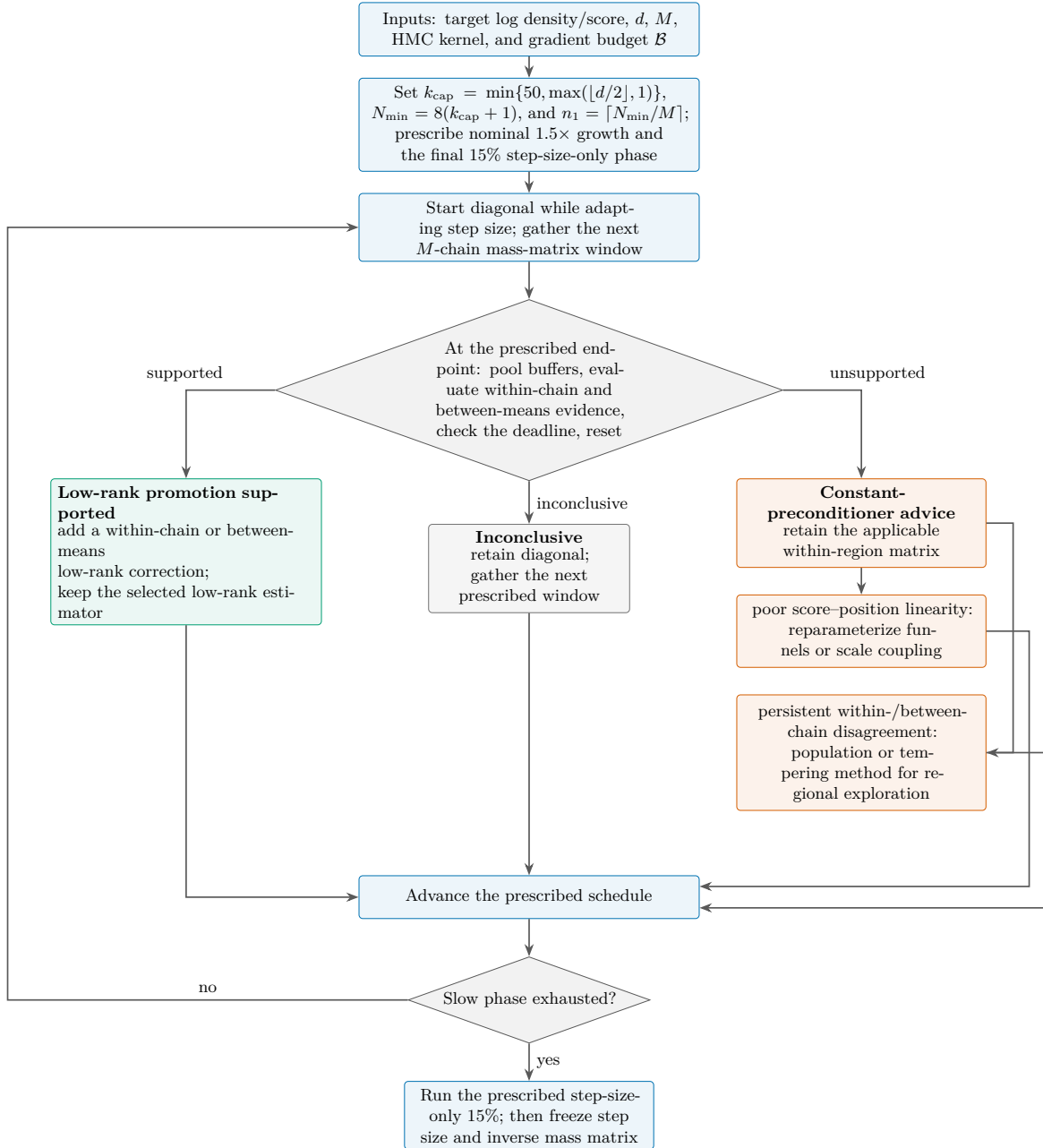


Figure 1: The evaluated automatic HMC warmup controller. Setup fixes every window boundary before execution, and evidence is evaluated only at mass-matrix-window endpoints. Every outcome continues the prescribed schedule: a supported decision promotes to a within-chain or between-means low-rank correction, inconclusive evidence retains the diagonal matrix for the next window, and an advisory outcome retains the applicable within-region inverse mass matrix and reports the advice supported by its signal. Without a supported promotion, the final matrix remains diagonal. Exhaustion of the mass-matrix-adapting phase enters the final step-size-only 15%, after which both step size and inverse mass matrix are frozen for posterior sampling; window boundaries remain fixed throughout.

4 Route-indexed attractor dynamics

This section analyzes confidence-calibrated, no-demotion selection among constant Euclidean inverse-mass estimators during HMC warmup. A “route” is the selected estimator, and the formal “latch” is the rule that prevents demotion during its route episode.

Let η_k be the target-population routing object at the current preconditioner, including the population moments and whitening transformation declared by the candidate estimator. Let C_k be its confidence set at the end of window k and H_b the eligibility region for promotion to route b . Coverage of C_k must pay for starting-law, within-window adaptation, whitening, sampling, and regularization errors. Routes are ordered and permit one transition.

Lemma 1 (Simultaneous soundness of the confidence-aware latch). *Let $\mathcal{K}$ be the fixed, declared set of potential decision windows. Suppose*

$$\Pr(\eta_k \notin C_k \mid \mathcal{F}_{k-1}) \leq \delta_k$$

for every $k \in \mathcal{K}$, whether or not a promotion is ultimately attempted. Promote to b only when $C_k \subset H_b$. Then, with probability at least $1 - \sum_{k \in \mathcal{K}} \delta_k$, every promotion made by the latch is supported by the true routing estimand at the time of promotion.

The lemma supports the declared target-population estimand at promotion time; persistence and global target coverage are separate questions. Repeated-window use requires confidence spending or an anytime-valid construction rather than reusing a fixed-window Wald set.

After promotion, consider a route episode for estimator a . Let $s(\theta)$ be its population routing statistic, g the router, $h_a(\theta)$ the population update field, and $\theta_a^* = \log G_a^*$. Write

$$F_a(\theta) = \theta + \alpha h_a(\theta), \quad \theta_{k+1} = \widehat{F}_{k,a}(\theta_k). \quad (2)$$

Theorem 2 (Local robust attractor of a route episode). *Let Θ be the bounded log-SPD chart used by the controller. Assume $R_0 > 0$, $\theta_a^* \in \Theta$, and $\overline{B}_F(\theta_a^*, R_0) \subset \Theta$, where $\overline{B}_F$ denotes a closed Frobenius-norm ball. Assume also that $s(\theta_a^*) \in \text{int}(\mathcal{S}_a)$, where $\mathcal{S}_a = \{s : g(s) = a\}$. Suppose $h_a(\theta_a^*) = 0$ and, on $\overline{B}_F(\theta_a^*, R_0)$, for constants $\mu > 0$ and $L_h > 0$,*

$$\langle \theta - \theta_a^*, h_a(\theta) \rangle_F \leq -\mu \|\theta - \theta_a^*\|_F^2, \quad \|h_a(\theta)\|_F \leq L_h \|\theta - \theta_a^*\|_F.$$

Let $0 < \alpha < 2\mu/L_h^2$ and $q = (1 - 2\alpha\mu + \alpha^2 L_h^2)^{1/2} < 1$. For every $k < K$, suppose candidate statistics and updates are constructed before routing, and define the joint error event

$$\begin{aligned} \mathcal{E}_k = \{ & \|\widehat{s}_k - s(\theta_k)\|_{\mathcal{S}} > r_k^s \} \\ & \cup \{ \|\widehat{F}_{k,a}(\theta_k) - F_a(\theta_k)\|_F > e_k \}, \quad \Pr(\mathcal{E}_k \mid \mathcal{F}_{k-1}) \leq \delta_k. \end{aligned}$$

Here $\|\cdot\|_{\mathcal{S}}$ is the declared statistic-space norm. Let s be L_s -Lipschitz from the ball to that norm, let $\kappa = \text{dist}\{s(\theta_a^), \partial\mathcal{S}_a\} > 0$, with distance in the same norm. Subject to the ordered latch, the observable rule selects or promotes estimator a whenever the following inclusion holds, and otherwise abstains:*

$$\overline{B}_{\mathcal{S}}(\widehat{s}_k, r_k^s) \subset \mathcal{S}_a.$$

For every $k < K$, assume $L_s R_0 + 2r_k^s < \kappa$ and $e_k \leq (1 - q)R_0$; the derived margin is sufficient for the observable inclusion on the good event. If $\|\theta_0 - \theta_a^\|_F \leq R_0$ and the latch is unset or*

already correctly set to a , then on the simultaneous good event the router selects a , the iterates remain in the ball, and

$$\|\theta_K - \theta_a^*\|_F \leq q^K \|\theta_0 - \theta_a^*\|_F + \sum_{j=0}^{K-1} q^{K-1-j} e_j. \quad (3)$$

This event has probability at least $1 - \sum_{k < K} \delta_k$.

The finite-window update-error budget is deliberately explicit:

$$e_k \leq e_k^{\text{start}} + e_k^{\text{step}} + e_k^{\text{white}} + e_k^{\text{sample}} + e_k^{\text{reg}}. \quad (4)$$

These terms cover the starting law, adaptive step size, estimated whitening, sampling fluctuation, and regularization. Controlled-Markov stability, convergence, and ergodicity results provide asymptotic ingredients under conditions such as drift, containment, and diminishing adaptation; they do not themselves provide route-specific finite-window high-probability bounds for e_k^{start} or e_k^{step} . Invariant kernels alone do not remove transient or state-dependent adaptation bias [Andrieu et al., 2005, Fort et al., 2016, Roberts and Rosenthal, 2007]. This theory therefore analyzes a confidence-calibrated estimator-selection rule. The evaluated controller uses fixed diagnostic thresholds; connecting it to the theorem requires calibration of the five component bounds in (4).

5 Uncertainty and safe structural decisions

5.1 Posterior covariance and estimator identification

For M equal-length chains with n recorded states, write $\bar{x}_m$ for chain m 's mean and $\bar{x}$ for the grand mean, and define

$$W = \frac{1}{M(n-1)} \sum_{m=1}^M \sum_{t=1}^n (x_{mt} - \bar{x}_m)(x_{mt} - \bar{x}_m)^\top,$$

$$B = \frac{n}{M-1} \sum_{m=1}^M (\bar{x}_m - \bar{x})(\bar{x}_m - \bar{x})^\top.$$

If S is the ordinary covariance of all Mn states about $\bar{x}$, then

$$(Mn-1)S = M(n-1)W + (M-1)B. \quad (5)$$

This finite-transcript identity is exact: it requires neither independence nor stationarity. It remains exact after one common realized linear projection A , with S, W, B replaced by $ASA^\top, AWA^\top, ABA^\top$. A projection selected from the same data is still algebraically descriptive, but calibrated post-selection inference requires cross-fitting, selective inference, or a simultaneous operator bound. Separately whitening S, W , and B does not preserve (5).

Now hold M fixed and let $n \rightarrow \infty$. Suppose each chain's empirical first and second moments converge to those of a law ν_m . Then S converges to the covariance of the equal aggregate mixture

$$\bar{\nu} = \frac{1}{M} \sum_{m=1}^M \nu_m.$$

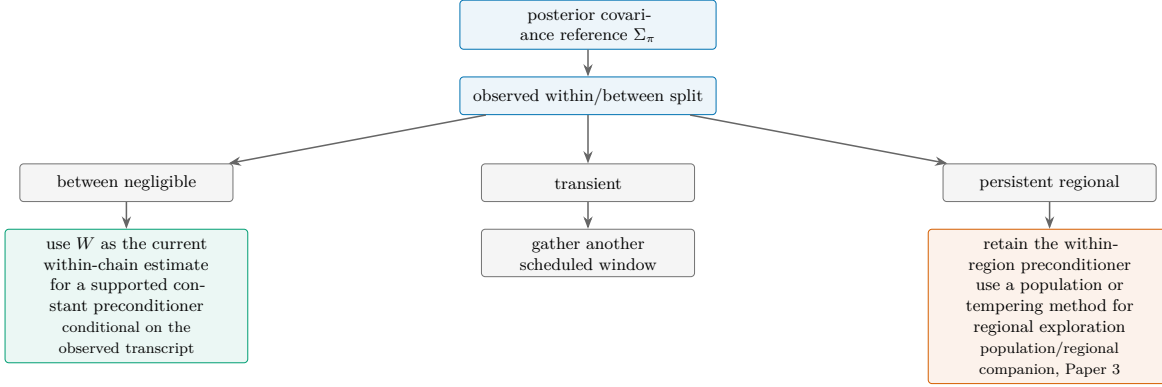


Figure 2: Practical interpretation of the posterior covariance reference through the observed within/between split. The leaves condition use of a constant Euclidean preconditioner on the observed transcript and identify the next action.

This limit identifies Σ_π only when $\bar{\nu}$ is second-moment representative of π . Agreement among chains is not enough: a clean ensemble confined to one basin can converge to the wrong regional covariance.

Proposition 3 (When covariance identifies an estimator). *Let $\mathcal{P}$ be a declared target class with finite covariance. An estimator functional T_a is identified by covariance on $\mathcal{P}$ if and only if it is constant on covariance fibres:*

$$\text{Cov}_{\pi_0}(X) = \text{Cov}_{\pi_1}(X) \implies T_a(\pi_0) = T_a(\pi_1) \quad (\pi_0, \pi_1 \in \mathcal{P}).$$

Equivalently, there is a map F_a on the covariance range of $\mathcal{P}$ such that $T_a(\pi) = F_a(\Sigma_\pi)$. This condition preserves the estimator-indexed target $G_a^ = T_a(\pi)$ rather than replacing it by one common preconditioner.*

The Fisher estimator gives a useful full-rank special case. Let $s(x) = \nabla \log \pi(x)$ and $J_\pi = E_\pi\{s(X)s(X)^\top\}$.

Proposition 4 (Full-rank Fisher covariance bound). *Suppose π has a smooth positive density, finite SPD covariance Σ_π , square-integrable score, and Stein boundary conditions. Then*

$$E_\pi\{(X - \mu)s(X)^\top\} = -I, \quad J_\pi \succeq \Sigma_\pi^{-1}.$$

For the unregularized, no-cutoff, full-rank affine-invariant Fisher population functional, with $\#$ denoting the SPD geometric mean,

$$G_F = \Sigma_\pi \# J_\pi^{-1}, \quad \Sigma_\pi^{-1/2} G_F \Sigma_\pi^{-1/2} = (\Sigma_\pi^{1/2} J_\pi \Sigma_\pi^{1/2})^{-1/2} \preceq I.$$

Equality holds precisely in the affine-score, hence Gaussian, case under these assumptions [Dembo et al., 1991].

This proposition does not identify the deployed finite-rank, masked, regularized Fisher estimator from covariance. For example, a standard Gaussian and a variance-one Student- t_ν law have the same covariance, whereas their one-dimensional location Fisher informations are respectively 1 and $\nu(\nu+1)/\{(\nu-2)(\nu+3)\} > 1$ for $\nu > 2$. Their full-rank Fisher functionals therefore differ.

5.2 A per-estimator Markov central limit theorem

For estimator a , collect its quadratic and score observables in a vector $f_a(X_t)$, with mean η_a . For a stationary, geometrically ergodic frozen-kernel chain with sufficient moments, the multivariate Markov-chain CLT gives

$$\sqrt{n}(\hat{\eta}_a - \eta_a) \Rightarrow \mathcal{N}(0, \Omega_a), \quad \Omega_a = \sum_{h=-\infty}^{\infty} \text{Cov}_{\pi}\{f_a(X_0), f_a(X_h)\}. \quad (6)$$

The full long-run covariance (LRV) depends on the functionals in f_a ; a single scalar ESS cannot be substituted into an iid Wishart law [Jones, 2004, Vats et al., 2018]. If ϕ_a maps moments to a symmetric residual, let J_a be the derivative of $\text{vech} \circ \phi_a$ and let

$$u = \text{vech}(\hat{R}_a - R_a),$$

where vech retains one copy of each off-diagonal. The delta method gives root- n limit covariance $\Gamma_a = J_a \Omega_a J_a^\top$ for $\sqrt{n}u$. Let Q be the oracle population support: its orthonormal columns span $\text{range}(\Gamma_a)$. Set $z = Q^\top u$ and require $p = \text{rank}(\Gamma_a) \geq 1$. If Γ_a is singular, assume $(I - QQ^\top)u = o_p(n^{-1/2})$, or that the residual lies exactly on the declared support. Require $Q^\top \hat{\Gamma}_a Q$ to be positive definite and consistent on this support. For one fixed window, an illustrative Wald construction is

$$r_{\alpha,n} = \left\{ \frac{2\chi_{p,1-\alpha}^2 \lambda_{\max}(Q^\top \hat{\Gamma}_a Q)}{n} \right\}^{1/2}. \quad (7)$$

Indeed, for symmetric A , $\|A\|_F \leq \sqrt{2}\|\text{vech}(A)\|_2$, which accounts for the factor 2 inside the squared radius and yields a Frobenius, hence operator, radius asymptotically. The construction requires a consistent multivariate LRV estimator and regularity for the estimator map. Estimating Q or its rank from the same window requires a separate justification. With M independent frozen chains, their contributions may scale as $1/M$; chains coupled through shared adaptation are not automatically independent. Non-asymptotic matrix concentration is another option only under explicit boundedness, tail, and dependence assumptions [Neeman et al., 2024].

Theorem 5 (Consequences of an operator-confidence event). *Let R_a and $\hat{R}_a$ be symmetric population and estimated residuals, with eigenvalues in descending order, and suppose $\|\hat{R}_a - R_a\|_{\text{op}} \leq r$.*

1. *Weyl's inequality gives $|\hat{\lambda}_j - \lambda_j| \leq r$ for every j .*
2. *For a declared population cutoff τ and $1 \leq k < d$, if $\hat{\lambda}_k - r > \tau$ and $\hat{\lambda}_{k+1} + r < \tau$, then exactly k population eigenvalues exceed τ . Otherwise this interval test is inconclusive.*
3. *Let V and $\hat{V}$ span the matched population and estimated top- k eigenspaces (or a declared isolated eigenvalue cluster), and let Δ be the minimum population separation between that cluster and its complement. If $\Delta > 2r$, then $\|\sin \Theta(\hat{V}, V)\|_{\text{op}} \leq 2r/\Delta$.*
4. *If $\lambda_{\min}(R_a) \geq m > r$, then*

$$\|\log \hat{R}_a - \log R_a\|_{\text{op}} \leq \log \left(\frac{m}{m-r} \right) \leq \frac{r}{m-r}.$$

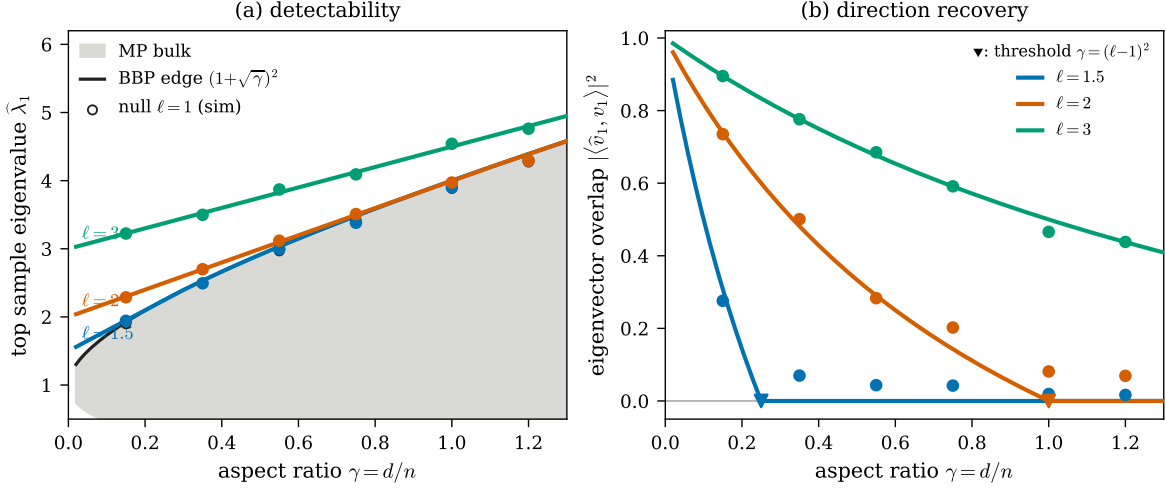


Figure 3: iid Gaussian spiked-PCA calibration ($d = 200$). The curves show the asymptotic sample eigenvalue and eigenvector overlap; markers are fixed-seed simulations. The population threshold is $1 + \sqrt{\gamma}$, and the null sample edge is $(1 + \sqrt{\gamma})^2$. The figure is not a diagnostic threshold for Markov-chain draws.

A preconditioner-error claim additionally requires a per-estimator perturbation bound for the moment-to-preconditioner map T_a .

These are standard operator, subspace, and SPD-log consequences [Bhatia, 1997, Yu et al., 2015, Higham, 2008]. They show exactly which margins a confidence-aware threshold test needs. The deployed cosine statistic Ψ is a deterministic agreement check between replicated directions. High agreement is compatible with shared whitening bias and does not establish accuracy or coverage.

5.3 The restricted role of BBP calibration

For iid Gaussian observations from the spiked covariance model with aspect ratio $\gamma = d/n$, a population spike separates when $\ell > 1 + \sqrt{\gamma}$, while the null sample-noise edge is $(1 + \sqrt{\gamma})^2$ [Baik et al., 2005, Baik and Silverstein, 2006, Paul, 2007]. This asymptotic PCA calculation is a useful scale calibration. It is not an exact edge for Markov draws, a finite-sample if-and-only-if statement, or an unrestricted information limit. In particular, subthreshold alternatives can retain nontrivial testing power [Onatski et al., 2013].

5.4 Controlled GMM at fixed marginal geometry

The controlled target is a five-dimensional, two-component Gaussian mixture with weights $\pi_1 = 0.30$, $\pi_2 = 0.70$, $v = (1, 1, 0, 0, 0)^\top / \sqrt{2}$, and $a = 21$. Its means and shared component covariance are

$$\mu_1 = -\pi_2 \delta v, \quad \mu_2 = \pi_1 \delta v, \quad \Sigma_w = I_5 + \beta v v^\top,$$

where

$$\delta^2(\text{SR}) = \frac{\text{SR}^2(1 + a)}{1 + \pi_1 \pi_2 \text{SR}^2}, \quad \beta = a - \pi_1 \pi_2 \delta^2.$$

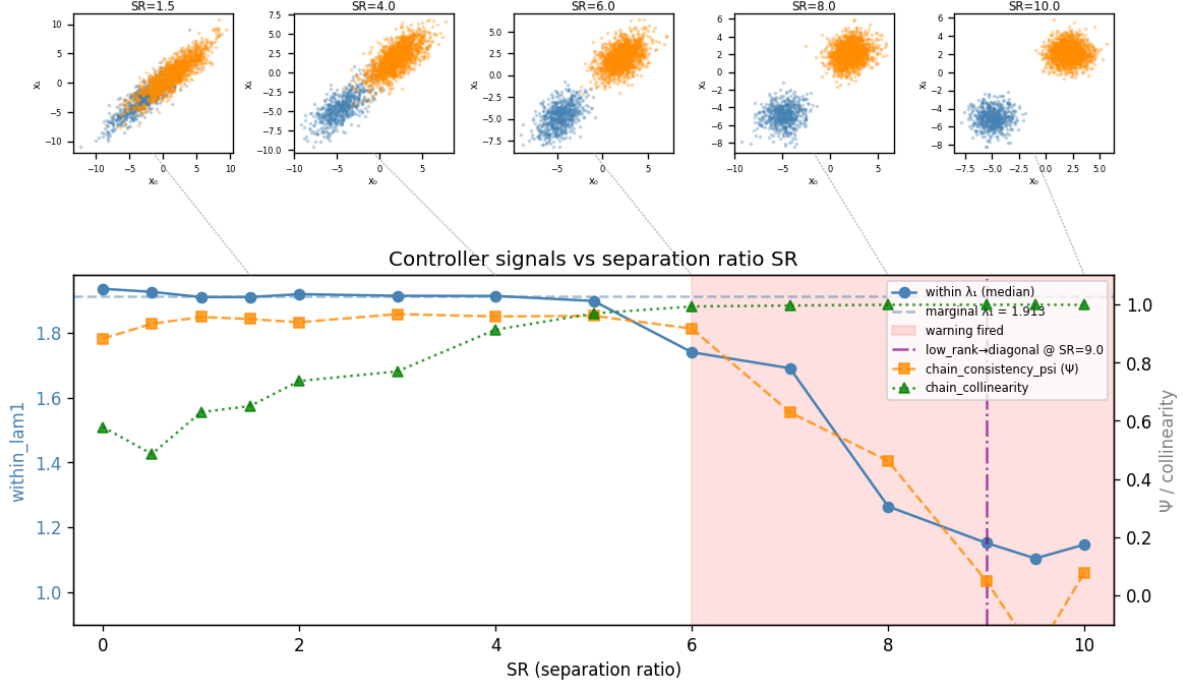


Figure 4: Current-corpus controlled GMM boundary, generated from the current $k = 2$ main corpus. Top: target draws at selected separation ratios. Bottom: seed-median current diagnostics. The marginal spike remains 1.913; the shading begins at the first grid point with a persistent disagreement signal in any seed, and the dash-dot line marks the first all-diagonal grid point. These are current-window signals, so a population handoff can clear.

Thus $\Sigma_{\text{marginal}} = \Sigma_w + \pi_1\pi_2(\mu_2 - \mu_1)(\mu_2 - \mu_1)^\top = I_5 + avv^\top$, so the marginal diagonal-whitened top eigenvalue is 1.913 throughout the separation sweep. Separation is indexed by $\text{SR} = \delta/s(\delta)$, where δ is the distance between component means and $s(\delta) = \{v^\top \Sigma_w v\}^{1/2}$ is the within-component standard deviation along the separation direction v . Equation (5) therefore gives an exact finite-transcript partition for this target. With latent component labels, the usual within-component plus label-mean-scatter identity gives the corresponding population partition.

The current sweep keeps the marginal spike fixed at 1.913 while the recorded within/between evidence changes. The selected estimator moves increasingly from low rank to diagonal, and an advisory population handoff appears. This associates the estimator-selection boundary with the changing within/between evidence rather than the fixed marginal spectrum; the recorded summaries leave the occupation mechanism unresolved. The current-window handoff is non-monotone. In every row, `global_exploration` remains `not_established`.

Compute changes when the disagreement becomes visible: the median onset is 5.0 at 20,000, 6.0 at 60,000, and 7.0 at 120,000 warmup gradients. Appendix C gives every current 60k row, the single-chain contrast, and the matched ablations.

6 Finite-horizon local transcripts

Structural evidence from visited states cannot by itself certify unseen regions. The appropriate obstruction is finite-horizon and depends on the joint event that every query remains local.

Theorem 6 (Local-transcript indistinguishability). *Let q_0, q_1 be smooth, positive, integrable unnormalized targets. Target access is only through a declared oracle, and the two targets give identical oracle responses on an open region A . Run the same randomized algorithm through a fixed horizon T from the same initial law under both targets, couple all random numbers, and assume it makes finitely many oracle calls almost surely through T . Let E_T be the event that every oracle query made by the complete, possibly interacting algorithm through T lies in A . Induction over oracle queries then gives identical coupled transcripts on E_T and a common submeasure of mass*

$$p_T = P_0(E_T) = P_1(E_T).$$

If the normalized targets π_0, π_1 have preconditioner functionals satisfying $d\{T(\pi_0), T(\pi_1)\} > 2\epsilon$, then every common transcript-measurable estimator $\hat{T}$ satisfies

$$\Pr_0\{d(\hat{T}, T(\pi_0)) > \epsilon\} + \Pr_1\{d(\hat{T}, T(\pi_1)) > \epsilon\} \geq p_T. \quad (8)$$

Consequently the two-point minimax error is at least $p_T/2$.

For replica events $E_{T,m}$, the chain rule is

$$p_T = \prod_{m=1}^M P(E_{T,m} \mid E_{T,1} \cap \cdots \cap E_{T,m-1}).$$

Independent replicas give $p_T = \prod_m q_{T,m}$. If every displayed conditional probability is bounded below by q_T , then $p_T \geq q_T^M$. Identical independent replicas are sufficient for $p_T = q_T^M$ when each has confinement probability q_T , but they are not necessary: equality in the displayed lower bound occurs whenever every conditional factor equals q_T , including some dependent constructions. Shared adaptation keeps the joint p_T form. The lower bound applies when $M = 1$ and makes multiple chains neither mathematically necessary nor sufficient for discrimination. Genuinely overdispersed multiple chains remain a principal practical safeguard against metastability and unvisited modes. Longer runs, more starts, or a more diverse collection of starts can reduce the confinement event, but need not do so at fixed compute. No fixed sample count is privileged: its meaning is target-, kernel-, start-, and controller-specific unless p_T is bounded for a declared problem class. This is a finite-horizon Le Cam two-point coupling argument [Yu, 1997].

7 Preconditioner selection and efficiency evidence

The headline NUTS benchmarks pair repeated seeded runs of a synthetic ill-conditioned Gaussian with a real-data German-credit Bayesian logistic-regression posterior. The latter is used as a posterior-geometry benchmark and carries no causal interpretation. The implemented controller selected low rank in all 12/12 evaluated headline runs.

Table 1 reports observed pooled ESS-per-gradient performance for automatic warmup relative to the prespecified Fisher low-rank warmup and Welford diagonal warmup configurations.

These roles and the nominal allocation were preregistered rather than chosen after observing the results. Both pooled baselines used $M = 8$ chains, 312 warmup transitions per chain, and the same proportional growing-window schedule. They differed in the mass-matrix estimator: Fisher low rank versus Welford diagonal. The runs used float64 arithmetic and explicit integer seeds. Pooled accounting divides the effective sample size computed from all eight chains by all warmup and sampling gradients. Every ratio charges gradients actually consumed.

Table 1: Geometric-mean pooled ESS-per-gradient ratios under the declared configurations. Ratio = automatic controller divided by prespecified baseline; values greater than 1 favor the automatic controller.

Model	prespecified warmup baseline	pooled ESS/gradient ratio
ill-conditioned Gaussian	Fisher low-rank warmup	2.451
ill-conditioned Gaussian	Welford diagonal warmup	22.572
German-credit logistic regression	Fisher low-rank warmup	1.951
German-credit logistic regression	Welford diagonal warmup	6.264

Across automatic, Fisher, and diagonal configurations, all 36/36 post-sampling pooled-chain quality checks passed: ArviZ rank-normalized split- $\hat{R}$ [Vehtari et al., 2021] was finite with maximum at most 1.01. The runs recorded 4,996 divergences during warmup and zero during post-warmup sampling.

These fixed benchmarks test efficiency after automatic selection. The controlled GMM sweep in Section 5 provides the complementary selection test: within/between evidence changes while the marginal geometry is held fixed.

Appendix B reports two secondary estimands. The marginal-amortized estimand treats one chain as the output and charges that chain’s sampling gradients plus one eighth of the jointly incurred warmup gradients; its geometric means aggregate paired seed-by-chain ratios. The one-output estimand also treats one chain as the output but charges all joint warmup gradients. Its comparator is the historical single-chain 2,500-warmup policy, so that comparison is descriptive and not budget-identical.

When M chains share one step size, the M acceptance statistics observed at a single step size are replicated measurements for one controller time, not M successive controller times. The shared-step comparison contrasts the current mean-pooled bundle with revision 2f62921848a93e7dc544ba9de8e29ef177e373b6, which applied those statistics as successive dual-averaging updates. The frozen historical/current ratios mean that the current bundle used 19.38–19.89 times fewer warmup gradients in the ill-conditioned runs and 31.82–35.60 times fewer for German credit. Mean pooling, the warmup tree-depth cap, and the controller semantics changed together, so these ratios describe the complete current-versus-historical configuration comparison.

In the other evaluated HMC-family kernels, low-rank selection occurred in 12/12 fixed-length and multinomial HMC runs with zero post-warmup divergences. Multinomial HMC lost to its comparator in 2/6 runs. Some fixed-length ratios have near-zero manual denominators, so the individual observed ratios are reported without a general performance summary.

8 Open-source implementation

BlackJAX provides the composable inference framework used here [Cabezas et al., 2024]. The controller is available in the `blackjax/adaptation/meta` module at merged revision `2103a4275b4d29b1650ba06458d5703eb7302b2e`.¹ The frozen experiment corpus was executed from pre-merge revision `29d2468857be4de1644ca4470c2a4aa7f8137656`; its `blackjax/adaptation/meta` tree is identical at the merged revision, and the frozen manifests retain the executed revision as the immutable run identity. The experiment code, frozen summary, checksummed outputs, and raw sidecar are archived with the paper in Project Geodesic.² `staged_adaptation` is the host interface that supplies the step-size controller, static schedule, and mass-matrix estimator.

```
1 warmup = staged_adaptation(  
2     blackjax.nuts, logdensity_fn,  
3     metric="auto",  
4     max_grad_budget=50_000,  
5     n_chains=8,  
6 )  
7 (last_state, parameters), info = warmup.run(rng_key, initial_positions)
```

The public interface exposes one compute budget and the chain population; Appendix E records the full signature.

9 Related work

Controlled-Markov stochastic approximation and adaptive-MCMC theory supply ingredients for stability, containment, transient control, and diminishing adaptation [Andrieu et al., 2005, Fort et al., 2016, Roberts and Rosenthal, 2007]. They do not by themselves establish the five-term finite-window update budget in (4). Markov CLTs and multivariate output analysis make LRV uncertainty functional-specific [Jones, 2004, Vats et al., 2018]. Matrix concentration under dependence is available under stronger boundedness and dependence hypotheses [Neeman et al., 2024].

Window adaptation [Carpenter et al., 2017] estimates a constant Euclidean mass matrix within a fixed schedule. Seyboldt et al. [2026] introduce a Fisher-divergence estimator of a constant linear or affine preconditioner from warmup draws and scores, with diagonal, dense, and low-rank-plus-diagonal forms, together with their own warmup schedule. In that method, mass-matrix adaptation and dual-averaging step-size adaptation remain separate. Entropy-based adaptive HMC is another approach to mass-matrix adaptation [Hirt et al., 2021]. Hird and Livingstone [2025] characterize when constant linear preconditioning can and cannot improve conditioning. Pathfinder instead builds local Gaussian approximations along an L-BFGS path for variational inference and initialization [Zhang et al., 2022]. ChEES tunes HMC trajectory length [Hoffman et al., 2021], while MEADS adapts an ensemble of generalized HMC chains [Hoffman and Sountsov, 2022].

Bou-Rabee et al. [2025] use “local adaptation” for changing the leapfrog step size within an HMC orbit in response to position-dependent curvature. Our controller instead adapts

¹<https://github.com/blackjax-devs/blackjax/tree/2103a4275b4d29b1650ba06458d5703eb7302b2e/blackjax/adaptation/meta>

²<https://github.com/arcueil/project-geodesic/tree/paper1-circulation-v1/papers/01-universal-warmup-path/experiments>

between warmup windows and freezes one constant Euclidean inverse mass matrix and one step size before posterior sampling; the methods address within-orbit and between-window adaptation, respectively.

For regional exploration, sequential Monte Carlo supplies population-level reweighting, resampling, and mutation over a sequence of bridging distributions [Del Moral et al., 2006]; this does not make resampling alone a guarantee of mode crossing. Tempered transitions are a direct regional/tempering precedent [Neal, 1996]. The affine-invariant ensemble sampler is included as an example of robustness to affine scaling, not as evidence of separated-mode exploration [Goodman and Weare, 2010].

Query-complexity lower bounds ask how many oracle calls are required to produce a total-variation-accurate sample over declared smoothness classes [Chewi et al., 2022, He and Zhang, 2025]. The stochastic-gradient, strongly-log-concave case gives a related oracle lower bound [Chatterji et al., 2022]. Those results differ from Theorem 6, which lower-bounds estimation of a target functional from a finite local transcript through its confinement mass p_T .

The corpus reports rank-normalized split- $\hat{R}$ in the sense of Vehtari et al. [2021]. Nested- $\hat{R}$ addresses a distinct design: superchains share an initializer within groups and are conditionally independent [Margossian et al., 2025]. It does not validate an arbitrary post-hoc grouped rank-normalized statistic, and no such extension is used here.

The iid calibration uses spiked-PCA theory [Baik et al., 2005, Baik and Silverstein, 2006, Paul, 2007], while Onatski et al. [2013] cautions against turning the PCA transition into an unrestricted testing impossibility. Weyl, Davis–Kahan variants, and matrix- function perturbation results support only the conditional threshold-test consequences stated in Theorem 5 [Bhatia, 1997, Yu et al., 2015, Higham, 2008].

10 Discussion

The deployed verdict reports the selected estimator and effective rank; its flags report conditioning scope, observed ensemble evidence, and advisory handoff. The next implementation step is to calibrate its fixed diagnostic thresholds to the theory’s uncertainty objects. Once our evidence-adequacy step establishes eligibility, the held-out criterion of Bales et al. [2019] could rank candidate metrics; we did not evaluate this composition.

The GMM boundary is compatible with the exact within/between decomposition but does not identify a unique occupation mechanism. Its $k = 3$ ablation exercises the deployed low-rank structure and finds no general efficiency gain. The finite-transcript result leaves both run length and starting diversity as ways to reduce the finite-horizon confinement probability p_T , subject to the actual interacting algorithm; it does not make multiple chains necessary.

Poor held-out score–position linearity directs the user toward reparameterization; persistent within-/between-chain disagreement directs the user toward a population or tempering method for regional exploration. Both are actions based on the observed warmup evidence rather than diagnoses of the posterior’s generating mechanism. Any selected constant inverse mass matrix remains available to an HMC method for within-region transitions. Fixed- L kernels use their declared integration length for gradient-budget accounting. The evaluated dimension/rank-capacity-indexed schedule keeps its decision boundaries fixed.

11 Conclusion

The Universal Warmup Path is an automatic warmup controller for the evaluated HMC family. It begins diagonal, evaluates evidence for each estimator at prescheduled endpoints, and either promotes to a supported low-rank-plus-diagonal inverse mass matrix or retains the diagonal matrix. It gathers another window when evidence is inconclusive. When the draws exhibit persistent within-/between-chain disagreement, they do not support treating one constant preconditioner as adequate across the observed regions. The controller retains the applicable within-region matrix and advises a population or tempering method for regional exploration. Poor held-out score–position linearity separately motivates reparameterization. After a supported estimator decision, the conditional convergence result bounds distance to that estimator’s population target by a contraction term plus accumulated finite-window estimation error. Operator uncertainty and local-transcript limits govern when that decision is available. In the evaluated headline benchmark NUTS runs, the controller achieved higher observed pooled ESS per gradient than both prespecified warmup baselines; all compared runs passed the declared post-warmup quality check. Schedule performance remained configuration-dependent. The method is scoped to between-window selection of a constant Euclidean inverse mass matrix and step size for HMC. The next step is calibration of estimator-selection uncertainty and integration with a population or tempering companion when persistent regional disagreement indicates a need for regional exploration.

Generative-AI use disclosure

Anthropic Claude Opus 4.8 and Claude Fable 5, and OpenAI Codex and GPT-5.6, supported literature discovery and checking, proof critique and derivation, code generation and review, experiment orchestration and validation, and manuscript drafting and editing. Junpeng Lao directed the work, reviewed and revised model output, checked model-supported claims against cited primary sources, source code, and frozen experiment outputs, and accepts responsibility for the manuscript. No AI system is an author or a source.

References

- Christophe Andrieu, Éric Moulines, and Pierre Priouret. Stability of stochastic approximation under verifiable conditions. *SIAM Journal on Control and Optimization*, 44(1):283–312, 2005. doi: 10.1137/S0363012902417267.
- Jinho Baik and Jack W. Silverstein. Eigenvalues of large sample covariance matrices of spiked population models. *Journal of Multivariate Analysis*, 97(6):1382–1408, 2006. doi: 10.1016/j.jmva.2005.08.003.
- Jinho Baik, Gérard Ben Arous, and Sandrine Péché. Phase transition of the largest eigenvalue for nonnull complex sample covariance matrices. *The Annals of Probability*, 33(5):1643–1697, 2005.
- Ben Bales, Arya Pourzanjani, Aki Vehtari, and Linda Petzold. Selecting the metric in Hamiltonian Monte Carlo. *arXiv preprint arXiv:1905.11916*, 2019. URL <https://arxiv.org/abs/1905.11916>.

- Rajendra Bhatia. *Matrix Analysis*. Springer, 1997.
- Nawaf Bou-Rabee, Bob Carpenter, Tore Selland Kleppe, and Sifan Liu. The within-orbit adaptive leapfrog no-U-turn sampler. *arXiv preprint arXiv:2506.18746*, 2025. URL <https://arxiv.org/abs/2506.18746>.
- Alberto Cabezas, Adrien Corenflos, Junpeng Lao, Rémi Louf, et al. BlackJAX: Composable Bayesian inference in JAX. *arXiv preprint arXiv:2402.10797*, 2024.
- Bob Carpenter, Andrew Gelman, Matthew D. Hoffman, Daniel Lee, Ben Goodrich, Michael Betancourt, Marcus Brubaker, Jiqiang Guo, Peter Li, and Allen Riddell. Stan: A probabilistic programming language. *Journal of Statistical Software*, 76(1):1–32, 2017. doi: 10.18637/jss.v076.i01.
- Niladri S. Chatterji, Peter L. Bartlett, and Philip M. Long. Oracle lower bounds for stochastic gradient sampling algorithms. *Bernoulli*, 28(2):1074–1092, 2022. doi: 10.3150/21-BEJ1377.
- Sinho Chewi, Patrik R. Gerber, Chen Lu, Thibaut Le Gouic, and Philippe Rigollet. The query complexity of sampling from strongly log-concave distributions in one dimension. In *Proceedings of the 35th Conference on Learning Theory*, volume 178 of *Proceedings of Machine Learning Research*, pages 2041–2059. PMLR, 2022. URL <https://proceedings.mlr.press/v178/chewi22b.html>.
- Pierre Del Moral, Arnaud Doucet, and Ajay Jasra. Sequential Monte Carlo samplers. *Journal of the Royal Statistical Society: Series B*, 68(3):411–436, 2006. doi: 10.1111/j.1467-9868.2006.00553.x.
- Amir Dembo, Thomas M. Cover, and Joy A. Thomas. Information theoretic inequalities. *IEEE Transactions on Information Theory*, 37(6):1501–1518, 1991. doi: 10.1109/18.104312.
- Simon Duane, A. D. Kennedy, Brian J. Pendleton, and Duncan Roweth. Hybrid Monte Carlo. *Physics Letters B*, 195(2):216–222, 1987.
- Gersende Fort, Éric Moulines, Amandine Schreck, and Matti Vihola. Convergence of Markovian stochastic approximation with discontinuous dynamics. *SIAM Journal on Control and Optimization*, 54(2):866–893, 2016. doi: 10.1137/140962723.
- Jonathan Goodman and Jonathan Weare. Ensemble samplers with affine invariance. *Communications in Applied Mathematics and Computational Science*, 5(1):65–80, 2010. doi: 10.2140/camcos.2010.5.65.
- Yuchen He and Chihao Zhang. On the query complexity of sampling from non-log-concave distributions (extended abstract). In *Proceedings of the 38th Conference on Learning Theory*, volume 291 of *Proceedings of Machine Learning Research*, pages 2786–2787. PMLR, 2025. URL <https://proceedings.mlr.press/v291/he25a.html>.
- Nicholas J. Higham. *Functions of Matrices: Theory and Computation*. SIAM, 2008. doi: 10.1137/1.9780898717778.
- Max Hird and Samuel Livingstone. Quantifying the effectiveness of linear preconditioning in Markov chain Monte Carlo. *Journal of Machine Learning Research*, 26(119):1–51, 2025.

- Marcel Hirt, Michalis K. Titsias, and Petros Dellaportas. Entropy-based adaptive Hamiltonian Monte Carlo. In *Advances in Neural Information Processing Systems*, volume 34, pages 28482–28495, 2021.
- Matthew D. Hoffman and Andrew Gelman. The no-U-turn sampler: Adaptively setting path lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research*, 15(47):1593–1623, 2014.
- Matthew D. Hoffman and Pavel Sountsov. Tuning-free generalized Hamiltonian Monte Carlo. In *Proceedings of the 25th International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 151 of *PMLR*, pages 7799–7813, 2022.
- Matthew D. Hoffman, Alexey Radul, and Pavel Sountsov. An adaptive-MCMC scheme for setting trajectory lengths in Hamiltonian Monte Carlo. In *Proceedings of the 24th International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 130 of *PMLR*, pages 3907–3915, 2021.
- Galin L. Jones. On the Markov chain central limit theorem. *Probability Surveys*, 1:299–320, 2004. doi: 10.1214/154957804100000051.
- Charles C. Margossian, Matthew D. Hoffman, Pavel Sountsov, Lionel Riou-Durand, Aki Vehtari, and Andrew Gelman. Nested $\hat{R}$: Assessing the convergence of Markov chain Monte Carlo when running many short chains. *Bayesian Analysis*, 20(4):1587–1614, 2025. doi: 10.1214/24-BA1453.
- Radford M. Neal. Sampling from multimodal distributions using tempered transitions. *Statistics and Computing*, 6(4):353–366, 1996. doi: 10.1007/BF00143556.
- Radford M. Neal. MCMC using Hamiltonian dynamics. In Steve Brooks, Andrew Gelman, Galin L. Jones, and Xiao-Li Meng, editors, *Handbook of Markov Chain Monte Carlo*, pages 113–162. Chapman and Hall/CRC, 2011.
- Joe Neeman, Bobby Shi, and Rachel Ward. Concentration inequalities for sums of Markov dependent random matrices. *Information and Inference: A Journal of the IMA*, 13(4), 2024. doi: 10.1093/imaiai/iaae032.
- Alexei Onatski, Marcelo J. Moreira, and Marc Hallin. Asymptotic power of sphericity tests for high-dimensional data. *The Annals of Statistics*, 41(3):1204–1231, 2013. doi: 10.1214/13-AOS1100.
- Debashis Paul. Asymptotics of sample eigenstructure for a large dimensional spiked covariance model. *Statistica Sinica*, 17(4):1617–1642, 2007.
- Gareth O. Roberts and Jeffrey S. Rosenthal. Coupling and ergodicity of adaptive Markov chain Monte Carlo algorithms. *Journal of Applied Probability*, 44(2):458–475, 2007. doi: 10.1239/jap/1183667414.
- Adrian Seyboldt, Eliot L. Carlson, and Bob Carpenter. Preconditioning Hamiltonian Monte Carlo by minimizing Fisher divergence. *arXiv preprint arXiv:2603.18845*, 2026. URL <https://arxiv.org/abs/2603.18845>.

- Stan Development Team. MCMC sampling using Hamiltonian Monte Carlo: Warmup schedule configuration. CmdStan User’s Guide, version 2.39, 2026. URL https://mc-stan.org/docs/2_39/cmdstan-guide-2_39.pdf. Accessed 2026-07-26.
- Dootika Vats, James M. Flegal, and Galin L. Jones. Strong consistency of multivariate spectral variance estimators in Markov chain Monte Carlo. *Bernoulli*, 24(3):1860–1909, 2018. doi: 10.3150/16-BEJ914.
- Aki Vehtari, Andrew Gelman, Daniel Simpson, Bob Carpenter, and Paul-Christian Bürkner. Rank-normalization, folding, and localization: An improved $\hat{R}$ for assessing convergence of MCMC. *Bayesian Analysis*, 16(2):667–718, 2021. doi: 10.1214/20-BA1221.
- Bin Yu. Assouad, fano, and Le Cam. In *Festschrift for Lucien Le Cam*, pages 423–435. Springer, 1997. doi: 10.1007/978-1-4612-1880-7_29.
- Yi Yu, Tengyao Wang, and Richard J. Samworth. A useful variant of the Davis–Kahan theorem for statisticians. *Biometrika*, 102(2):315–323, 2015. doi: 10.1093/biomet/asv008.
- Lu Zhang, Bob Carpenter, Andrew Gelman, and Aki Vehtari. Pathfinder: Parallel quasi-Newton variational inference. *Journal of Machine Learning Research*, 23(306):1–49, 2022.

A Proofs and technical details

Proof of Lemma 1. On the event $\eta_k \in C_k$, the inclusion $C_k \subset H_b$ implies $\eta_k \in H_b$. Iterated expectation and a union bound over the fixed potential-window set $\mathcal{K}$ give $\Pr(\exists k \in \mathcal{K} : \eta_k \notin C_k) \leq \sum_{k \in \mathcal{K}} \delta_k$. This bound is taken before observing which windows trigger an opportunity. Every realized promotion is therefore supported on the complementary event. $\square$

Proof of Proposition 3. If $T_a = F_a \circ \text{Cov}$, equal covariance plainly implies equal estimator target. Conversely, if T_a is constant on every covariance fibre, define $F_a(\Sigma)$ to be $T_a(\pi)$ for any $\pi \in \mathcal{P}$ with $\text{Cov}_\pi(X) = \Sigma$. Fibre constancy makes this definition independent of the representative. $\square$

Proof of Proposition 4. Integration by parts under the Stein boundary conditions gives $E\{(X - \mu)s(X)^\top\} = -I$. Hence the covariance matrix of $(X - \mu, s(X))$ is

$$\begin{pmatrix} \Sigma_\pi & -I \\ -I & J_\pi \end{pmatrix} \succeq 0.$$

Its Schur complement gives $J_\pi \succeq \Sigma_\pi^{-1}$. The definition of the SPD geometric mean then yields

$$\Sigma_\pi^{-1/2}(\Sigma_\pi \# J_\pi^{-1})\Sigma_\pi^{-1/2} = (\Sigma_\pi^{1/2} J_\pi \Sigma_\pi^{1/2})^{-1/2} \preceq I.$$

Equality in the Schur complement makes $s(X) = -\Sigma_\pi^{-1}(X - \mu)$ almost surely. Integrating this affine score identifies the Gaussian density; the Gaussian attains equality. For the variance-one Student- t_ν example, direct integration of the squared location score gives $J_\pi = \nu(\nu + 1)/\{(\nu - 2)(\nu + 3)\}$. $\square$

Proof of Theorem 2. Write $d_k = \theta_k - \theta_a^*$. Dissipativity and the growth bound give

$$\|d_k + \alpha h_a(\theta_k)\|_F^2 \leq (1 - 2\alpha\mu + \alpha^2 L_h^2) \|d_k\|_F^2.$$

For any $z \in \overline{B}_S(\hat{s}_k, r_k^s)$, the route inequalities give

$$\|z - s(\theta_a^*)\|_S \leq 2r_k^s + L_s \|\theta_k - \theta_a^*\|_F < \kappa.$$

Because $s(\theta_a^*)$ is interior to $\mathcal{S}_a$, this proves $\overline{B}_S(\hat{s}_k, r_k^s) \subset \mathcal{S}_a$. The candidate update is therefore selected without conditioning its error event on the route. Equation (2) then gives $\|d_{k+1}\|_F \leq q\|d_k\|_F + e_k$. The basin condition closes the induction, iteration yields (3), and a conditional union bound gives the stated probability. $\square$

Proof of Theorem 5. The eigenvalue intervals are Weyl’s inequality. The two strict inequalities around τ certify exactly k eigenvalues above the cutoff. Applying the Davis–Kahan sin- Θ theorem on an event with gap exceeding $2r$ gives the displayed conservative bound. Finally, the spectral floor keeps both matrices SPD along the perturbation path; the integral representation of the matrix logarithm bounds its derivative by the reciprocal spectral floor, giving $\log\{m/(m-r)\} \leq r/(m-r)$. None of these steps specifies the map from the residual moments to an estimator’s preconditioner, hence the separate T_a condition. $\square$

Proof of Theorem 6. The initial states and randomized choices are coupled. If the first $j-1$ queries lie in A , their oracle responses agree under q_0 and q_1 , so the algorithm states and the j th query agree. Induction over all oracle calls therefore makes the complete transcripts identical on E_T and proves the common submeasure of mass p_T . On that submeasure the estimator output is the same. The two ϵ -balls around $T(\pi_0)$ and $T(\pi_1)$ are disjoint, so that output must be wrong for at least one target. Integrating over the common mass yields (8); taking the larger error gives the minimax bound. $\square$

B Secondary efficiency estimands

The marginal-amortized estimand divides one chain’s effective sample size by that chain’s sampling gradients plus one eighth of the warmup gradients shared by the eight chains. It uses the same pooled Fisher low-rank warmup and Welford diagonal warmup baselines described in Section 7. The one-output estimand divides one chain’s effective sample size by that chain’s sampling gradients plus all shared warmup gradients. It compares with historical single-chain versions of those estimator families and is descriptive rather than budget-identical.

Table 2: Secondary geometric-mean ESS-per-gradient ratios. Ratio = automatic controller divided by the listed comparator; values greater than 1 favor the automatic controller. The one-output comparison is descriptive and not budget-identical.

Model	warmup baseline family	marginal-amortized	one-output
ill-conditioned	Fisher low-rank warmup	2.324	1.409
ill-conditioned	Welford diagonal warmup	21.124	14.196
German credit	Fisher low-rank warmup	1.882	1.131
German credit	Welford diagonal warmup	5.887	6.009

C Controlled GMM boundary study

The target construction pins the marginal diagonal-whitened top eigenvalue at 1.913. Here $SR = \delta/s(\delta)$ is the separation ratio defined in §5. The main 60k sweep is:

Table 3: Current 60k $k = 2$ sweep across three seeds. “Persistent” counts `persistent_disagreement_signal`; “handoff” counts the advisory `population` endpoint.

SR	low-rank/diagonal	persistent	handoff	within λ_1 range
0	3/0	0/3	0/3	1.910–1.979
0.5	3/0	0/3	0/3	1.916–1.932
1	3/0	0/3	0/3	1.907–1.952
1.5	3/0	0/3	0/3	1.907–1.962
2	3/0	0/3	0/3	1.895–1.937
3	3/0	0/3	0/3	1.882–1.929
4	3/0	0/3	0/3	1.913–1.920
5	3/0	0/3	0/3	1.842–1.907
6	3/0	2/3	2/3	1.737–1.847
7	3/0	3/3	2/3	1.371–1.707
8	2/1	2/3	2/3	1.205–1.266
9	0/3	2/3	2/3	1.093–1.297
9.5	0/3	2/3	2/3	1.102–1.258
10	1/2	2/3	2/3	1.081–1.219

Single-chain contrast. At the evaluated budget and seed, one chain selects low rank at $SR = 1.5$, diagonal at $SR = 5$, and diagonal at $SR = 10$. The last row has mode-weight estimate 1.0. Ensemble evidence is unassessed for these single-chain rows. The outcomes are specific to the implemented diagnostic thresholds and evaluated budget.

Matched diagonal ablation. The comparison holds warmup, step size, initialization, and $M = 8$ fixed. Its low-rank/matched-diagonal projection bulk $ESS(v^\top x)$ per gradient ratios are 0.869230, 0.997721, 1.170288, and 0.997752 at $SR = 1.5, 4, 5, 9$, respectively. These four evaluated comparisons are descriptive: identification of anisotropy alone does not establish a general efficiency gain.

Three-axis main sweep. For $k = 3$, all three seeds report persistent disagreement from the first evaluated $SR = 6$ grid point onward. The endpoint population handoff is nevertheless 2/3 at each of those separation ratios and can clear within a run. The persistent evidence field and current-window handoff therefore answer different questions.

Three-axis matched ablation (null). A same- M paired ablation used $k = 3$ correlated axes³ at $SR \in \{3.0, 3.5, 4.0, 4.5\}$, seeds 1001–1016, and $M = 8$. Both arms shared the 60k

³For v supported equally on k axes, diagonal whitening gives $\lambda_1(k) = (1 + a)/(1 + a/k)$. For $k = 2, \dots, 5$, the marginal spikes are $\{1.913, 2.750, 3.520, 4.231\}$ and the off-diagonal counts are $\{1, 3, 6, 10\}$, respectively. For $k = 2$, the diagnostic-threshold implementation’s asymptotic covariance-spike comparison is $1.913 < 2$, where 2 is its escalation threshold; this is neither a general impossibility nor a finite-budget guarantee. At $k = 3$, $2.750 > 2$, so that particular threshold contrast is absent.

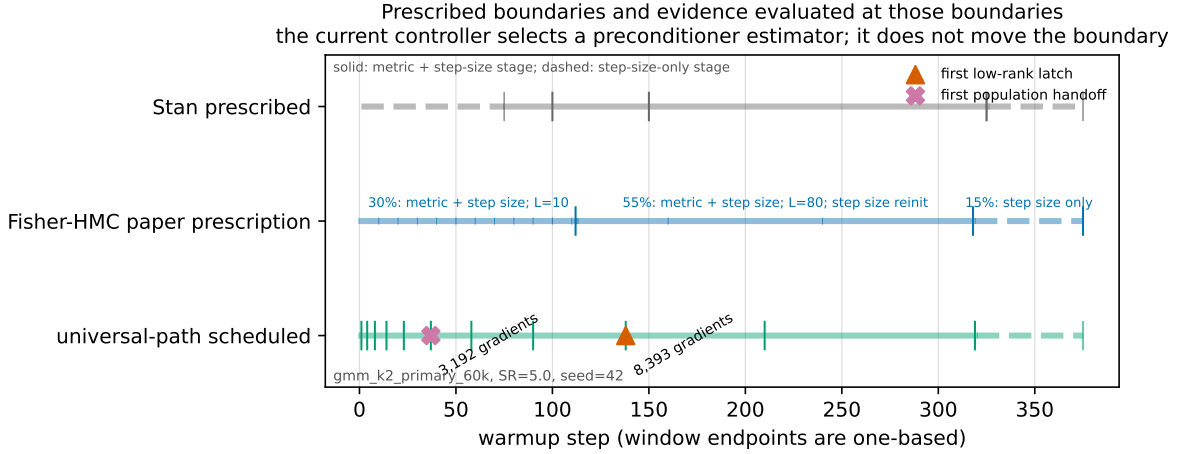


Figure 5: Schedule evidence and one executed controller trace. The Stan and Fisher-HMC lanes are unexecuted reference prescriptions; Fisher-HMC denotes the specific 30/55/15 prescription in Seyboldt et al. [2026]. Controller estimator decisions are evaluated at prescribed boundaries, which remain fixed throughout. In the displayed trace the first population-handoff marker is transient and clears.

warmup, terminal state, and step size; each then produced 4,000 draws per chain. For equal-SR, within-seed pairs, the low-rank/matched-diagonal projection bulk $\text{ESS}(v^\top x)$ per gradient geometric-mean ratio was 1.037; the seed-cluster t 95% interval was $[0.998, 1.077]$. Because the interval includes 1, the stated measurable, non-noise go/no-go criterion fails: this is a null result and supports no efficiency claim. Per-SR geometric means were 1.002, 0.999, 0.987, and 1.169 in ascending SR order; the $\text{SR} = 4.5$ rise is local and does not generalize. All 64/64 runs completed, with 16/16 valid low-rank profiles at each SR, zero divergences, and maximum projection split- $\hat{R}$ 1.006. Of the 64 paired ratios, 39 are above 1 and 25 below.

D Schedule-comparison scope

Full factorial. The validated schedule-by-buffer experiment records poor radon performance for the *bundle* combining proportional-growing windows with an accumulating/per-draw-recompute buffer. Because accumulation and recomputation move together in this arm, the observation estimates performance of that complete bundle. The remaining configurations show the joint performance of schedule, buffer, and controller semantics. The controller boundaries in Figure 5 are prescribed before execution and remain fixed through the run.

Restart ablation. Continuous dual averaging has mixed efficiency: it wins 3/6 evaluated runs, with continuous/reseed ratios from 0.9464 to 1.1385 and geometric means 1.014 for the ill-conditioned target and 1.024 for German credit. It reduces the recorded oscillation and number of warmup divergences; these are the observed configuration-level results of the limited reseed ablation.

E Full warmup signature

```
1 staged_adaptation(  
2     algorithm, logdensity_fn, metric="welford_diag" | MetricRecipe | MetricCore,  
3     *, max_grad_budget=None, n_chains=1, imm_shrinkage_to_previous=0.0,  
4     initial_inverse_mass_matrix=None, initial_step_size=1.0,  
5     target_acceptance_rate=0.80, adaptation_info_fn=..., integrator=velocity_verlet,  
6     schedule_fn=None, initial_metric_state=None, **extra_parameters)
```

Step-size dual averaging and the stage schedule live in the host; mass-matrix estimation is delegated to `metric`.